

# AMARNATH S. PATEL

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## EDUCATION

**University of Central Florida** **Undergraduate Student** **4.00 GPA**

B.S. in Physics (Optics & Lasers) and Mathematics, Computer Science Minor (38 credit hours) August 2025 – Present

**Florida Atlantic University** **3.66 GPA**

University coursework completed via FAU High School, ages 14–18 (111 credit hours) August 2021 – May 2025

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## RESEARCH EXPERIENCE

**UCF Astrophotonics Lab, CREOL — Undergraduate Researcher** August 2025 – Present

*Supervisor: Dr. Stephen Eikenberry*

- Contribute to a measurement campaign characterizing the wavelength-dependent complex transfer matrix of a photonic lantern by [off-axis digital holography](#) collect and analyze holographic data across all-port × C-band (1525–1575 nm) sweeps, reaching ≈ 98% reconstruction fidelity (extending the method of Dobias et al., *Opt. Express* 2026).
- Developed and validated the phase-retrieval analysis: FFT sideband isolation and demodulation, Butterworth filtering, and numerically optimized quadratic-phase correction, followed by LP-mode decomposition to recover complex modal amplitude and phase.
- Automated the four-instrument measurement chain — tunable IR laser, motorized polarization control, and [IR beam-profiling camera](#) with real-time analysis and FITS output — for unattended overnight acquisition.
- Contribute to [PolyOculus](#), an astronomical instrument program building photometric observing capability from a networked array of 8 telescopes: developed INDI-protocol mount control, focuser automation, RA/Dec slewing, and backlash compensation.

**UCF Physics Department — Undergraduate Research Assistant** March 2026 – Present

*Physics Education Research — Dr. Zhongzhou Chen (NSF Award 2421299; Gates Foundation INV-076932)*

- Research on automated assessment generation for introductory physics instruction: built [ESTELA](#), a system producing multi-version isomorphic exams from structured problem banks, and an agentic generation–evaluation pipeline that iteratively drafts and validates new problems against instructor criteria.
- Authored the documentation and distribution materials used at a faculty workshop, and maintain the public release pipeline that external adopters depend on.
- Lead developer across two independent implementations of the assessment tool, including technical ownership of a parallel version authored by another group member. Manuscript in preparation.

**FAU Grant-Funded AI Safety Research Project (High School)** January 2024 – March 2025

*Supervisor: Tucker Hindle, Florida Atlantic University*

- Benchmarked 10+ coherence and detection methods (GPT-2 perplexity, RoBERTa, BERT NSP, LSA, NLI, burstiness) across 3,125+ generated sequences; identified GPT-2 perplexity as the strongest discriminator (3.3× separation between coherent and incoherent text).
  - Characterized a consistent quality–evasion tradeoff across four adversarial humanization pipelines. Grant-funded with HPC access; presented at the Wilkes Honors College Symposium (2025).
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## OTHER EXPERIENCE

**Secretary — National Society of Black Physicists (NSBP), UCF Chapter** 2025 – Present

- Coordinate meetings, records, and communications for the UCF student chapter of the National Society of Black Physicists.

**President — Coding Club, Florida Atlantic University (High School)**

2023 – 2025

- Founded and led a student coding club; organized meetings and collaborative programming projects with peers.

**Teaching Assistant — Calculus (High School)**

August 2024 – May 2025

- Assisted 70 undergraduate students with learning calculus; office hours, exam review, grading. Part-time (10 h/week).

**Advanced Experimental Vehicles — Programmer, Leader, Builder (High School)** November 2023 – May 2025

- Configured Arch Linux ARM on Raspberry Pi 5 with Hyprland compositor and WireGuard VPN, enabling worldwide real-time telemetry monitoring of BMS, GPS, and camera feeds.
  - Won 2nd Place in Division and Lockheed Martin Award for “Highest Level of Engineering Excellence.”
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**PRESENTATIONS****Complex Transfer-Matrix Characterization of a Photonic Lantern by Off-Axis Digital Holography** November 2026 (upcoming)

Joint Annual Conference of the National Society of Black Physicists and the National Society of Hispanic Physicists, North Carolina.

**Batch Processing for Automated Grading via Azure OpenAI**

June 2026

University of Central Florida, Downtown Campus. ESTELA project (Physics Education Research, Dr. Zhongzhou Chen).

**Coherence and Detection Approaches for Identifying AI-Generated Text**

March 2025

Wilkes Honors College Undergraduate Research Symposium, Florida Atlantic University.

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**TECHNICAL REPORTS****Coherence and Detection Approaches for Identifying AI-Generated Text**

2025

- Empirical study of AI-text detection and adversarial evasion: benchmarked 10 coherence and detection methods (GPT-2 perplexity best,  $3.35\times$  separation; RoBERTa the only effective detector) and built four evasion pipelines, characterizing a consistent quality–evasion tradeoff. Grant-funded, Florida Atlantic University.
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**SELECTED PROJECTS****CELERIS — Metalens Design Pipeline & From-Scratch RCWA Solver**

2026

- From-scratch solver for Maxwell’s equations in periodic subwavelength structures via rigorous coupled-wave analysis / the Fourier modal method: 1D TE/TM multilayer gratings with Li/Liu–Fan inverse-rule factorization, full 2D-vectorial  $P \cdot Q$  formulation for bi-periodic nanopillar arrays, and stable Redheffer scattering-matrix layer recursion.
  - Validated every layer of the stack against closed-form physics, an independent method, or energy conservation: RCWA vs. transfer-matrix method to  $1e-6$ , multilayer S-matrix to  $1e-15$ , 2D→1D degenerate reduction to  $5e-12$ ,  $\Sigma$  (diffraction efficiency) = 1.000000, and diffraction-limited designed focal spots at  $\text{FWHM} = \lambda f / D$ . Independently cross-validated against the grcwa and Stanford  $S^4$  solvers ( $\approx 1e-7$ ) and reproduces canonical published metalenses (Khorasaninejad et al., *Science* 2016; Chen et al., *Nat. Nanotechnol.* 2018).
  - Implemented the physical design and analysis capability standalone RCWA kernels lack: Sellmeier and tabulated  $n, k$  dispersion, meta-atom library sweeps with automatic etch-depth selection for full- $2\pi$  phase coverage, gradient-based (Adam) inverse design, Pancharatnam–Berry geometric-phase and polarization-multiplexed lenses exploiting form birefringence (reporting per-polarization RMS phase and focal isolation in dB), plus Strehl, Zernike wavefront, MTF, chromatic focal shift, through-focus caustic, and fabrication-tolerance Monte-Carlo analysis with fabrication-ready GDSII export.
  - Accelerated Rayleigh–Sommerfeld far-field propagation as a CUDA kernel:  $675\times$  over a 16-core CPU at 92k pillars ( $23.9\text{ s} \rightarrow 35\text{ ms}$ ), agreeing to  $\approx 1e-5$ , with automatic CPU fallback. C++23/CUDA, with Python bindings and a native desktop GUI. *Sole-author manuscript in preparation (Computer Physics Communications).*
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TECHNICAL SKILLS

**Programming Languages:** C/C++, CUDA, Rust, Python, JavaScript, Shell (Fish, Bash, tcsh), LaTeX/Typst

**Scientific Computing:** NumPy, SciPy, GPU/CUDA acceleration, FFT & signal processing, electromagnetic simulation (RCWA)

**Tools & Libraries:** Qt/PySide6, Docker, Git, CMake, XMake, INDI, KaTeX

**Instrumentation:** GPIB/VISA, RS-232, GigE Vision camera acquisition, tunable IR laser control, motorized polarization control, FITS & GDSII data formats

**Operating Systems:** Linux (Arch, NixOS, Ubuntu), BSD, Windows

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RELEVANT COURSEWORK

University of Central Florida

|                                                       |                                              |
|-------------------------------------------------------|----------------------------------------------|
| <i>Physics</i>                                        |                                              |
| Geometric Optics & Lab                                | Modern Physics [PHY 3101]                    |
| Quantum Information Processing                        | Mathematical Methods for Physics [PHZ 3113]  |
| Independent Research [PHY 4912] (In Progress)         | Electricity & Magnetism I (In Progress)      |
| <i>Mathematics</i>                                    |                                              |
| Applied Linear Algebra [MAS 3105]                     | Complex Analysis [MAA 4402]                  |
| Linear Algebra (Proof-Based) [MAS 3106] (In Progress) | Partial Differential Equations (In Progress) |
| <i>Computer Science</i>                               |                                              |
| C Programming [EGN 3211]                              | Discrete Structures                          |
| Object-Oriented Programming (In Progress)             |                                              |

Florida Atlantic University

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|--------------------------------------|--------------------------------------------|
| <i>Physics</i>                       |                                            |
| General Physics I                    | General Physics II — Honors                |
| <i>Mathematics</i>                   |                                            |
| Calculus I                           | Calculus II                                |
| Honors Calculus III                  | Ordinary Differential Equations [MAP 2302] |
| Elementary Matrix Algebra [MAS 2103] |                                            |
| <i>Computer Science</i>              |                                            |
| Data Structures & Algorithms         | Computer Logic Design                      |
| Computer Architecture                | Deep Learning [CAP 4613]                   |
| C++ Programming [COP 3014]           | Web Development [COP 3813]                 |

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HONORS & AWARDS

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|--------------------------------------------------------------------------------------------|------|
| 1st Place, Night Hacks Hackathon — UniUtils (course schedule generator & classroom finder) | 2023 |
| Lockheed Martin Award — “Highest Level of Engineering Excellence,” AEV Competition         | 2024 |
| 2nd Place in Division, Advanced Experimental Vehicles Competition                          | 2024 |
| Florida Bright Futures — Florida Academic Scholars (highest tier; 100% tuition)            | 2025 |
| Barry M. Goldwater Scholarship — UCF institutional nominee                                 | 2026 |
| Astronaut Scholarship Foundation — UCF institutional nominee                               | 2026 |